Elbaraa Refaie

Senior Automation Engineer | Packaging & EOL Systems

□ +905072164613 | □ elbaraarefaie202015@gmail.com | □LINKEDIN PROFILE (Click) | □ Portfilo : https://elbaraarefaie.github.io/

Summary

Senior Automation Engineer | Packaging & End-of-Line Systems | Schneider PacDrive Specialist

Results-driven Automation Engineer with 4+ years of end-to-end ownership of EOL packaging machinery — from PLC architecture and servo motion programming through FAT, international commissioning, after-sales support, and customer-driven upgrades. Designed, commissioned, and actively supports 100+ machines for multinational FMCG manufacturers across 6 countries, maintaining 100–150 active support connections annually. Deep expertise in Schneider Electric PacDrive 3 ecosystem — LMC series controllers, MC12 MultiCarrier linear motor systems, multi-axis servo coordination, and reusable Function Block development including checkweigher integration. Architected and deployed an embedded two-layer Production Intelligence Platform (PacDrive 3 ST logic + Jmobile IPC reporting) running as standard across the Vervo machine portfolio — delivering OEE shift reporting with scheduled PDF export, machine state tracking with CSV diagnostics log, and a 24-hour production speed historian at 1-second resolution, daily production records over last month, hourly production records for 48 hours. Equivalent functionality to a dedicated MES and SCADA historian installation — at zero external software cost. Delivered flagship projects including a 550 PPM tray lid line (16+ servo axes, MC12, Schmalz vacuum, 12 glue guns) and a proprietary Smart Infeed System merging 5 lines at 400 PPM total throughput. Published researcher in machine vision and AI; B.Sc. Electrical-Mechatronics Engineering, CGPA 3.94/4.0.

Core Skills

- ***PLC & Motion Control:*** *Schneider PacDrive 3, LMC106/212/216, LMC 802/902 Pro, MC12 MultiCarrier, EcoStruxure Machine Expert, IEC 61131-3 Structured Text, Function Blocks, multi-axis servo motion control*
- ***HMI & Production Data:*** *Exor Jmobile, Vijeo Designer, OEE monitoring, production reporting, alarm management, SCADA concepts, OPC-UA*
- ***Industrial Networks:*** *CANopen, Ethernet IP, Modbus TCP, Sercos, fieldbus communication*
- ***Commissioning & Support:*** *FAT / SAT, machine commissioning, remote diagnostics, after-sales support, format changeover, technical training, technical documentation*
- ***Leadership:*** *Software team leadership, code standardization, reusable library development, cross-functional collaboration.*

Experience

AUTOMATION ENGINEER & SOFTWARE DEPARTMENT LEAD

Vervo for Advanced Packaging (EOL packaging machine builder), Istanbul, Turkey *NOV 2022 – Present*

Sole automation engineer responsible for the full programming lifecycle of EOL packaging machinery — from software architecture and PLC development through FAT, site commissioning, and post-sales support — serving major FMCG manufacturers across 6 countries.

- ***Lead the software department****; own software architecture, coding standards and delivery for every machine, and* ***train internal engineers, operators and after-sales staff*** *on PacDrive 3 operation, HMI use and troubleshooting.*
- *Programmed and commissioned* ***100+ EOL packaging machines*** *including box erectors, case packers, and tray lid systems for clients including Ülker, AdkuTurk (Indomie TR), Tadım, Torku, Tadım Germany, Meray Germany, Indomie Saudi Arabia & Serbia, Hayat Kimya, Eruslu, BIM, Mutlu Un, İpek Pamuk* and others
- *Designed and commissioned a* ***550 PPM high-speed*** *tray lid packaging line at Ülker Gebze — coordinating 16+ servo axes, 7 conveyors,* ***MC12 MultiCarrier****, Schmalz vacuum gripper, 12 hot-melt glue guns, and an automated reject module for length control and overlap detection, all on Schneider PacDrive 3 with Exor Jmobile HMI. For interpack Exhibition (7–13 May 2026, Düsseldorf),* ***Increased machine speed to 650 PPM*** *on the MC12 MultiCarrier platform, coordinating 16+ servo axes, 7 conveyors, 12 hot-melt glue guns, Schmalz vacuum gripper and an automated reject module.*
- *Developed the control system for a* ***double-entry bar-product eco machine (up to 30 boxes/min, 650 PPM) without the MC12 system****, lowering the machine hardware cost and making the product more competitively priced*
- Developed a proprietary Smart Infeed System (Akıllı Ürün Taşıma Sistemi) capable of dynamically merging 2–5 independent upstream production lines (each at 80 PPM) into a single synchronized 400 PPM feed for one case packing machine — eliminating the need for multiple case packers and significantly reducing client capital and floor space costs
- *Developed all machine software on* ***Schneider PacDrive 3 / LMC series*** *platform using* ***IEC 61131-3 Structured Text****, covering motion control, safety logic, recipe management, and fault diagnostics. This machine is running in* ***production at Protein Ocn.***
- *Designed and deployed* ***operator HMIs*** *with multilingual support,* reducing operator error rate by an estimated 30% and cutting average fault-resolution time through structured alarm management and multilingual interface design
- *Established* ***standardized reusable code libraries*** *for common machine functions,* reducing average PLC programming time per new machine by approximately 25%, enabling faster project delivery across repeat machine types
- *Led FAT and SAT for 100+ machines with full documentation delivery — including functional specs, parameter backups, and test protocols — across 6 countries: Turkey, Germany, Saudi Arabia, Serbia, Egypt, and Malaysia*
- Designed and deployed an embedded two-layer Production Intelligence Platform across the Vervo machine portfolio — PacDrive 3 Structured Text FB library handling all KPI calculation and state logic, with Exor Jmobile IPC managing storage, reporting, and visualization — comprising:
  - **OEE Reporting:** shift-configurable tracking of products, daily production record for 30 days, hourly records for 48 hours, rejections, boxes, stops, homing events, and stop time — with on-demand and scheduled PDF export from HMI
  - **Machine State Tracker:** continuous logging of all state transitions with time-in-state, product/box counters, and CSV export for remote fault diagnosis

- **Production Speed History:** 24-hour rolling trend at 1-second resolution — functioning as an embedded SCADA historian natively on the Jmobile IPC

- *Delivered **operator and maintenance training** to client engineering teams post-commissioning,* reducing first-month downtime incidents through structured handover training and commissioning documentation delivered to client maintenance teams.
- *Managed **full after-sales service lifecycle for 250+ installed machines** — including fault diagnosis, software corrections, parameter recovery, and emergency remote intervention. Maintained an average of **100–150 active machine support connections per year** across installed base — providing consistent technical coverage across Turkey, Germany, Arabia, Serbia and Egypt*

***Key Technologies:** Schneider PacDrive 3, LMC106/212,216, LMC 802/902 Pro, MC12 MultiCarrier, EcoStruxure Machine Expert, Structured Text (ST), Servo Motion Control, HMI (Exor Jmobile)*

BUSINESS DATA ANALYST

STUDYFANS COMPANY, Istanbul, Turkey *JUN 2022 – OCT 2022*

- *Automated repetitive reporting workflows using Python, reducing manual data processing time by an estimated 60%*
- *Built Tableau dashboards tracking KPIs for department heads and documented standard workflows.*

AUTOMATION & DIGITALIZATION INTERN

TOP GLOVE CORPORATION, Malaysia *JULY 2021 – OCT 2021*

- *Implemented real-time sensor dashboards for bulk tank monitoring and built inventory applications with Microsoft Power Apps.*
- *Collaborated with cross-functional automation teams on live production systems.*

RESEARCH ASSISTANT

CENTER OF ROBOTICS AND ARTIFICIAL INTELLIGENCE (CAIRO UTM) MARCH 2021 – JAN 2023

- *Co-authored 2 peer-reviewed publications in machine vision and AI-based traffic analysis (IOP Publishing, Springer).*
- *Developed computer vision systems for vehicle classification and counting using YOLO and DeepSORT, and a facial-gesture computer control system for amputee accessibility.*

Key Projects

1. ***High-Speed Bar Products Packaging Line (550 → 650 PPM)***
   - *Schneider PacDrive 3, LMC, MC12 MultiCarrier, EcoStruxure Machine Expert, Exor Jmobile HMI.*
   - *Synchronized 16+ servo axes, 7 conveyors and 12 glue guns timed to carrier position; reject logic combining length measurement and overlap detection; recipe management with single-operator changeover.*
2. ***Double-Entry Bar-Product Machine (650 PPM, 30 boxes/min) Cost-optimized, no MC12***
   - *Control system and HMI for a double-entry machine reaching 650 PPM on conventional servo architecture, reducing machine cost versus an MC12-based design.*
3. ***Smart Infeed System (400 PPM) Proprietary technology***
   - *Real-time servo speed synchronization of 2–5 lanes, gap correction before the case packer, and partial line shutdown without stopping the packer.*
4. ***Embedded Production Intelligence Platform***
   - *OEE reporting, machine state tracker, 24-hour speed historian (86,400 points/day), daily production record for 30 days, hourly records for 48 hours, production bar graphs and preventive-maintenance counters, with CSV export enabling remote fault diagnosis and fewer site visits.*

Publications

1. *Loon, Lee Pin, Elbaraa Refaie, and Ahmad Athif Mohd Faudzi. "Data analytics for traffic flow prediction in Custom using Long Short-Term Memory (LSTM) networks." Journal of Physics: Conference Series. Vol. 2107. No. 1. IOP Publishing, 2021. Retrieved from: https://doi.org/10.1088/1742-6596/2107/1/012006*

2. *Refaie, E., & Faudzi, A. A. M. (2022). Machine Vision-Based System for Vehicle Classification and Counting Using YOLO. In Computational Intelligence in Machine Learning (pp. 391-398). Springer, Singapore. Retrieved from: https://doi.org/10.1007/978-981-16-8484-5_37*

Education

B.Sc. Electrical Engineering (Mechatronics), Universiti Teknologi Malaysia (UTM) **Sep 2017 – Feb 2022**

- CGPA 3.94/4.0. Final Year Project (grade A+): Integrated traffic system for vehicles classification and counting and traffic prediction using DeepSORT and LSTM algorithms.

Mechanical Engineering, Zagazig University, Egypt **Sep 2014 – Apr 2016**

- Completed 3 semesters with 89% before pursuing Electrical-Mechatronics Engineering at UTM, Malaysia.

Languages

Arabic: Native | ***English: Professional*** | ***Turkish: Upper-intermediate***

References

Baris Menderes	*Vervo Development Manager*	*baris.menderes@vervo.com.tr*	
Prof. Dr. Zaharuddin Mohamed	*Head of Advanced Control Research Group, School of Electrical Engineering*	Contact details available upon request	Google Scholar
Prof. Ir. Ts. Dr. Ahmad 'Athif Bin Mohd Faudzi	*Director of Centre for Artificial Intelligence and Robotics (CAIRO UTM)*	Contact details available upon request	Google Scholar